Transition-risk core: how `L`, `Δp`, and the GVA shock

- `A` — technical coefficients (global MRIO)
- `L` — the Leontief inverse
- `Δp` — the carbon-tax price change, per sector
- `ΔV/V` — the relative GVA (value-added) shock, per sector → the paper's Table 4

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1. Inputs

Symbol	Meaning	Source
`Z`	inter-industry flow matrix (600×600)	`DATA_12R/ICIO2025_12R_2022.csv`, region-industry block
`x`	gross output per sector (600)	ICIO `OUT` row
`GVA`	value added per sector (600)	ICIO `VA` row
`CI`	direct carbon intensity, tCO_2e per USD-million output	`DATA_12R/CARBON_INTENSITY_12R_2022.csv`
`XCE`	carbon price, USD/tCO_2e	parameter = **70** (paper's Table 4 value)
`ϕ`	cost pass-through fraction	parameter, swept **0** → **1** in 10% steps

The index is 12 regions × 50 ISIC industries = **600 region-sectors** (labels like `GBR_C24A`). ROW is the accounting closure region and is kept in the matrix so the global IO system balances, but carries no financial interpretation.

Units. ICIO is in current USD millions; `CI` is tCO_2e per USD-million of output. With `XCE` in USD/t, the per-output carbon charge is a **dimensionless fraction of output value**:

$$ct_j = CI_j \times XCE \times 10^{-6}.$$

The 10^{-6} reconciles the units: the charge $CT_j = CI_j x_j \times XCE$ comes out in plain USD, while output x_j is in USD *millions*, so dividing the two requires the factor of a million.

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2. Step 1 — technical coefficients `A` and Leontief inverse `L`

Column-normalise the flow matrix by gross output (Eq. 2), then invert:

$$A = Z\,\hat{x}^{-1}, \qquad A_{ij} = \frac{Z_{ij}}{x_j}, \qquad L = (I - A)^{-1}.$$

Zero-output columns are guarded before the division.

`L` is the **demand-side** Leontief inverse; its column sums are sector output multipliers. Because ICIO is a true multi-regional table, `A` is the full global block matrix, so `L` embeds cross-border linkages automatically (a shock in one region propagates to all others via the off-diagonal trade blocks).

The price side uses the **transpose (dual)** form $\tilde{L} = (I - A^\top)^{-1}$ (Eq. 4).

A validity check: the spectral radius of `A` must be < 1 (productive economy). For this data it is ≈ **0.579**, so `(I − A)` is invertible.

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3. Step 2 — carbon charge vector `ct_direct`

Multiply each sector's carbon intensity by the carbon price (Eq. 6), aligning the carbon-intensity table (industries × regions) to the ICIO region-sector ordering:

$$\mathbf{ct}_{\text{direct}} = \mathbf{CI} \times XCE \times 10^{-6}.$$

This is the fraction of each sector's output value taken by the carbon tax on its **Scope-1** emissions. (Scope 2/3 are captured up/downstream through the IO linkages, so only direct emissions enter here.)

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4. Step 3 — price change `Δp` (modified Leontief dual, Eq. 8)

Only a fraction `ϕ` of the charge is passed into prices. With $\hat{\phi} = \mathrm{diag}(\phi)$, the **modified Leontief dual** and the resulting price change are:

$$\tilde{L}(\phi) = \left(I - A^\top \hat{\phi}\right)^{-1} \hat{\phi}, \qquad \Delta p = \tilde{L}(\phi)\, \mathbf{ct}_{\text{direct}}.$$

`Δp_j` is the relative increase in sector *j*'s output price. It is **0 at φ=0** (nothing passed on) and rises monotonically to full pass-through at φ=1.

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5. Step 4 — GVA shock `ΔV/V` (Eq. 10)

The carbon charge splits into a *price* part (`Δp`) and a *retained-in-GVA* part. Differencing the sector cost identity before/after the tax (Eqs. 3, 5, 9) under inelastic demand (final demand `y` fixed) gives the per-unit-output value-added change:

$$\Delta v = \left[(I - A^\top)\, \tilde{L}(\phi) - I + \hat{\phi}\right] \mathbf{ct}_{\text{direct}}.$$

Scale to absolute and then to a **relative** shock:

$$\Delta V_j = x_j\, \Delta v_j, \qquad \boxed{\frac{\Delta V_j}{V_j} = \frac{x_j\, \Delta v_j}{\mathrm{GVA}_j}}.$$

Sign convention. Negative = value added lost. `Δp` and the GVA shock are complementary: cost not passed into prices shows up as a GVA loss.

Sanity anchors (the paper's two special cases)

φ	`Δp`	GVA shock `ΔV/V`
0 (absorb all)	0	$-CT_j/\mathrm{GVA}_j$ (most negative)
1 (pass all on)	max	$+CT_j/\mathrm{GVA}_j$ (mirror image)

The script verifies the region-level version: UK shock is **−0.851%** at φ=0 and **+0.851%** at φ=1, matching $\mp CT_{\mathrm{UK}}/\mathrm{GDP}_{\mathrm{UK}}$.

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6. Step 5 — aggregating per-sector → per-nation

A region's GDP is the sum of its sectors' GVA, so the national relative shock is the total ΔGVA over total GVA — equivalently a **GDP-share-weighted average** of the per-sector shocks:

$$\frac{\Delta\mathrm{GDP}_r}{\mathrm{GDP}_r} = \frac{\sum_{j\in r}\Delta V_j}{\sum_{j\in r}\mathrm{GVA}_j} = \sum_{j\in r} f_{j,r}\,\frac{\Delta V_j}{V_j}, \qquad f_{j,r} = \frac{\mathrm{GVA}_j}{\sum_{k\in r}\mathrm{GVA}_k}.$$

The weighting scheme differs by quantity: the GVA/GDP shock is weighted by **GVA share**, whereas `Δp` is rolled up with **output weights**, since a producer-price index is output-weighted rather than value-added-weighted.

The per-sector `ΔV` already contains cross-border cascading, so a national number includes the drag from pricier imported inputs; aggregation just sums the region's own sectors.

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7. Outputs

File	Content
`out_gva_shock_by_region_phi.csv`	national GVA/GDP shock (%), 12 regions × φ
`out_price_change_by_region_phi.csv`	national output-weighted `Δp` (%), 12 regions × φ
`out_gva_shock_by_sector_phi.csv`	full **600 region-sectors × φ** GVA shock (%)
`out_price_change_by_sector_phi.csv`	full **600 region-sectors × φ** `Δp` (%)